Experiment 5 — Pretrained Vision Transformer on Colab

This notebook fine-tunes an ImageNet-pretrained ViT-B/16. It warms up the classification head and then unfreezes only the final two encoder blocks. Training a transformer from scratch is intentionally avoided.

1. Configuration

```
Mode   : Local
Device : NVIDIA RTX PRO 2000 Blackwell Generation Laptop GPU
```

2. Stage images and verify the shared split

	rows	classes
train	62312	27
validation	16984	27

3. Build ViT-compatible datasets

4. Warm up the head, then unfreeze the last two encoder blocks

	phase	phase_epoch	train_loss	validation_loss	accuracy	weighted_f1	macro_f1
0	head	1	1.9695	1.8602	0.5589	0.5521	0.5077

	phase	phase_epoch	train_loss	validation_loss	accuracy	weighted_f1	macro_f1
0	last_two_blocks	1	1.7608	1.7133	0.6100	0.6020	0.5606

	phase	phase_epoch	train_loss	validation_loss	accuracy	weighted_f1	macro_f1
0	last_two_blocks	2	1.5916	1.6791	0.6224	0.6206	0.5759

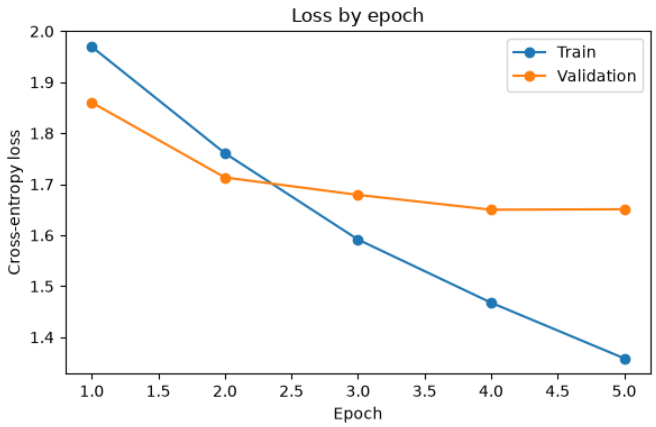
	phase	phase_epoch	train_loss	validation_loss	accuracy	weighted_f1	macro_f1
0	last_two_blocks	3	1.4678	1.6501	0.6402	0.6340	0.5927

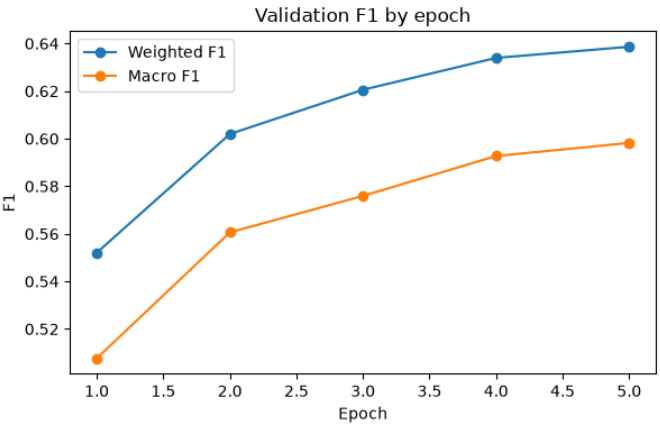
	phase	phase_epoch	train_loss	validation_loss	accuracy	weighted_f1	macro_f1
0	last_two_blocks	4	1.3586	1.6509	0.6404	0.6386	0.5982

5. Training curves

Use validation loss and macro F1 to assess generalisation. A falling training loss combined with a rising validation loss indicates overfitting.

	global_epoch	phase	phase_epoch	train_loss	validation_loss	accuracy	weighted_f1	macro_f1
0	1	head	1	1.9695	1.8602	0.5589	0.5521	0.5077
1	2	last_two_blocks	1	1.7608	1.7133	0.6100	0.6020	0.5606
2	3	last_two_blocks	2	1.5916	1.6791	0.6224	0.6206	0.5759
3	4	last_two_blocks	3	1.4678	1.6501	0.6402	0.6340	0.5927
4	5	last_two_blocks	4	1.3586	1.6509	0.6404	0.6386	0.5982





6. Evaluate and save probabilities

	validation_loss	accuracy	weighted_f1	macro_f1
vit_b16	1.6509	0.6404	0.6386	0.5982

	prdtypecode	precision	recall	f1	support
8	1281	0.326	0.343	0.334	414
6	1180	0.548	0.301	0.388	153
24	2585	0.503	0.365	0.423	499
7	1280	0.474	0.409	0.439	974
22	2582	0.438	0.459	0.449	518
2	50	0.509	0.408	0.453	336
26	2905	0.833	0.316	0.458	174
11	1302	0.470	0.512	0.490	498
17	2220	0.585	0.461	0.515	165
16	2060	0.423	0.682	0.522	999
0	10	0.485	0.637	0.551	623
1	40	0.520	0.616	0.564	502
12	1320	0.660	0.492	0.564	648
13	1560	0.746	0.507	0.604	1015
20	2462	0.657	0.581	0.617	284
4	1140	0.651	0.639	0.645	534
19	2403	0.779	0.563	0.654	955
10	1301	0.723	0.634	0.675	161
21	2522	0.660	0.722	0.690	998
15	1940	0.741	0.658	0.697	161
3	60	0.734	0.681	0.706	166
9	1300	0.681	0.751	0.714	1009
25	2705	0.747	0.690	0.718	552
18	2280	0.677	0.794	0.731	952
14	1920	0.846	0.791	0.818	861
23	2583	0.805	0.847	0.826	2042
5	1160	0.888	0.928	0.907	791

Interpretation

Compare ViT-B/16 with the strongest CNN on the identical validation products. Keep it only if the gain justifies its longer runtime and memory use. A transformer trained from scratch is not part of this experiment.